

Feature Spec: EPS Ceiling Proration for Loss of Pay, NCP Days in ECR, EDLI Base

Audience: Product, QA, Support

Module: Payroll → EPF / EPS / EDLI, ECR (EPFO challan) report

Status: Implemented, unit tested, ready for QA

1. Why this change

Every month the employer files an ECR (Electronic Challan cum Return) with EPFO. EPFO expects an employee's pension (EPS) wage to fit the days the employee actually contributed. Our rule caps it at:

$$\text{EPS wages} \leq \text{EPS ceiling} \times (\text{calendar days} - \text{NCP days}) / \text{calendar days}$$

EPFO's published ECR rules require that EPS wages do not exceed EPF wages and that EPS remitted does not exceed 8.33% of EPS wages. The NCP-based cap above is our interpretation (see section 9).

- **EPS ceiling** is ₹15,000 (₹25,000 from 17-Sep-2026, the confirmed effective date of the revision).
- **NCP days** (Non-Contributory Period) are the days in the month the employee was not paid. They reduce the employee's pensionable service.

Problems before this change

Problem	Effect
EPS wages were reported at the full ceiling even when the employee had loss of pay (LOP)	EPFO could reject the ECR line
NCP days were reported only for employees with EPF restricted and NC on	Most employees with LOP were filed with NCP 0, so pension service was over-stated
Employees with nothing paid were not reported with full-month NCP	ECR line did not reflect a non-contributory month
EDLI wage was tied to the EPS wage	Once the EPS ceiling is reduced for NCP days, EDLI would wrongly shrink too

2. What changes for the user

Area	Before	After
NCP days in ECR	Only restricted + NC employees	Every EPF member with LOP (restricted or not, NC on/off, EPS on/off, low earners)
Nothing paid in the month	NCP not reported	NCP = calendar days , all wages 0
EPS wage with LOP	Full ceiling	Ceiling reduced for NCP days

Area	Before	After
EDLI wage	Same as EPS wage	Lower of the EPF wage (after LOP) and the full ceiling ; the ceiling is not reduced for NCP days
Employer 12% total	—	Unchanged (only the EPS vs EPF A/c 1 split moves)
Employee PF and net pay	—	Unchanged
Additional earning days	—	Extra days worked in the month: count as paid days and reduce NCP

3. Glossary

Term	Meaning
Working days	Organisation working days in the pay period (e.g. 26)
Paid days	Working days – LOP days
Calendar days	Days in the wage month (28–31). Taken from the payroll date, so a 26th-to-25th cycle uses the month it is filed for
NCP days	Non-contributory days reported in ECR
EPS contributory calendar days	Calendar days – NCP days. A calendar-day count (not working days) used only to reduce the EPS ceiling
EPF wage	Wage on which EPF is computed (after restriction / NC rules)
EPS wage	Wage on which the 8.33% pension share is computed
EDLI wage	Wage on which the 0.5% EDLI is computed
A/c 1	Employer EPF share = employer 12% – EPS
NC setting	EPF setting for restricted employees: with NC on, the employee keeps the full capped EPF wage until their earned wage falls below the cap

4. Business rules

4.1 NCP days

effective paid days = paid days + additional earning days

if effective paid days = 0:

NCP = calendar days (nothing paid)

else:

NCP = working days – effective paid days (never below 0, rounded to a whole day)

NCP is capped at calendar days – 1 (at least one EPS contributory calendar day when anything is paid)

- **Exception, nothing paid:** when no wages are paid for the whole month, NCP = **calendar days** (e.g. 31 in October), not working days. In every other case NCP is counted in working days.
- NCP is rounded to the nearest whole day, halves rounded up (ECR accepts whole days only): 0.5 day of LOP gives NCP 1, 0.25 day gives NCP 0.

- NCP = LOP days from attendance **plus** working days before the date of joining or after the date of exit. Non-working days (weekly offs, holidays) are never paid or unpaid days, so they do not count as NCP, unless an unpaid leave is applied on them (it is then an LOP day).
- NCP means unpaid **working days** for payroll purposes, including working days before the date of joining or after the date of exit (wages are not due for them, EPFO's NCP definition). This matches how wages are paid (monthly wage $\times$ paid days $\div$ working days). Non-working calendar days are never NCP unless an unpaid leave is applied on them.

4.2 EPS wage

reduced EPS ceiling = ceiling $\times$ EPS contributory calendar days / calendar days (rounded down to the rupee)

EPS wage = lower of (employer EPF wage, reduced EPS ceiling)

EPS = 8.33% of EPS wage

EPF A/c 1 = employer 12% - EPS

- **Employer EPF wage** (the starting point for EPS and EDLI):
 - Unrestricted: actual EPF wage $\times$ paid days $\div$ working days (e.g. $78,266.50 \times 24/26 = 72,246$).
 - Restricted, NC off: cap $\times$ paid days $\div$ working days.
 - Restricted, NC on: lower of the cap and (actual EPF wage $\times$ paid days $\div$ working days).
 - **Additional earning days** are added **on top**: cap (or actual wage if unrestricted) $\times$ additional days $\div$ working days. This extra is contributed to employee PF, A/c 1 and admin; EPS stays at its ceiling and EDLI at the full ceiling.
- When EPS is not applicable (EPS off and not mandatory), EPS wage and EPS are 0 and the full 12% goes to A/c 1. NCP is still reported.
- The "low earner, EPS mandatory" check still uses the **full** monthly wage against the **full** ceiling. LOP never changes who is a mandatory EPS member.

4.3 EDLI wage

EDLI wage = lower of (employer EPF wage, full ceiling)

EDLI = 0.5% of EDLI wage

- The EPF wage is already prorated for LOP, so EDLI follows LOP through the wage.
- Unlike EPS, the **ceiling** is not reduced for NCP days: EDLI is insurance cover and is not linked to pensionable service days.

4.3a Admin charges

Admin = 0.5% of employer EPF wage – no ceiling

- Unchanged by this feature. Follows whatever wage the employer PF is paid on.

4.4 September 2026 (ceiling change on 17-Sep)

- **Each employed day uses the ceiling in force that day:** ₹15,000 for 1–16 Sep, ₹25,000 from 17 Sep. An employee who joins on or after 17 Sep is on ₹25,000 for every employed day (EPFO FAQ Q7 Scenario A); a leaver before 17 Sep is on ₹15,000 only.
- For a full-month employee the two parts weigh 16/30 and 14/30.

- **Restricted wage:** each part is capped at its own ceiling, then blended. Full month, no LOP: $15,000 \times 16/30 + 25,000 \times 14/30 = \mathbf{19,666.67}$, the full-month **September blended ceiling** (not a statutory ceiling).
 - Example, ₹20,000 restricted with NC on, 24 of 26 days paid: earned $20,000 \times 24/26 = 18,461.54$. 1–16 Sep: $\min(15,000, 18,461.54) = 15,000$. 17–30 Sep: $\min(25,000, 18,461.54) = 18,461.54$. Blended: $15,000 + 3,461.54 \times 14/30 = \mathbf{16,615.38}$.
- **EPS wage is worked out per part, then blended:** in each part, EPS wage = lower of (that part's EPF wage, that part's reduced ceiling); the two results are blended by employed days. The 4.2 formula therefore applies to each part separately in September, not to the month's blended totals.
- **Reduced EPS ceiling:** each part's ceiling is reduced for the same NCP days and **rounded down first**, then blended. NCP 2: $\text{floor}(15,000 \times 28/30) = 14,000$ and $\text{floor}(25,000 \times 28/30) = 23,333 \rightarrow 14,000 + 9,333 \times 14/30 = \mathbf{18,355.40}$ (slightly below $19,666.67 \times 28/30 = 18,355.56$ because of the per-part rounding).
- The blend follows EPFO FAQ Q7 (each ceiling applies to its own days). EPFO has said September portal validations may be relaxed, so the portal's exact September check is not confirmed.

4.5 Payruns calculated before this release

- Old payruns are **not** rewritten.
- If an old payrun with LOP still has EPS on the full ceiling, ECR reports it the way the earlier ECR did: **NCP 0** and the EPS wage on which EPS was remitted, so the line is not rejected and EPS wage matches EPS remitted. This is detected when:
 - the stored EPS wage is above the reduced EPS ceiling, or
 - no EPS wage is stored and the stored EPS contribution is above 8.33% of the reduced ceiling.
- For September 2026 the reduced ceiling is blended over the employee's employed days (as payroll does), so new September payruns, including joiners after 17-Sep, keep their real NCP.
- Old September 2026 payruns stored **without** EPS/EPF wage bases: ECR fills the bases from payroll but always reports the **recorded** contribution amounts (as deducted). If the recorded EPS is above 8.33% of the filled (NCP-reduced) EPS base, the row is treated as legacy: EPS wage on the unreduced base, NCP 0.
- Every such line is recorded in the server log with a note to recalculate the payrun before filing.
- **Recalculating** the payrun applies the new rules.
- **September 2026:** payruns calculated before this release should be recalculated before the September ECR is filed.

5. When will the EPS wage visibly go down?

The rule runs for everyone, but it only lowers the number when the employer EPF wage after LOP is still **above** the reduced ceiling.

Employer setup	EPS wage reduced with LOP?	Why
Restricted + NC on , wage above the cap	Yes	EPF wage stays at the cap, ceiling is reduced
Unrestricted employer, wage above the ceiling	Yes	EPF wage is the actual prorated wage, well above the ceiling
Restricted + NC off	Usually no	EPF wage already drops by $\text{paid} \div \text{working days}$, which is more than the ceiling reduction

Employer setup	EPS wage reduced with LOP?	Why
Low earner (wage ≤ ceiling)	Usually no	Same reason
Additional days ≥ LOP	No	NCP is 0

In all cases **NCP days are still reported** in ECR.

6. QA test plan

6.1 Setup

- September 2026 payrun, **26 working days**, payroll date in September (30 calendar days).
- For each case: set LOP → recalculate → check the payslip / statutory values → download the **September ECR** and check the employee's line.
- Cases 1–11 are on the payroll for the whole of September, so restricted wages use the September blended ceiling (19,666.67 for a full month). Case 12 joins on 21 Sep and uses ₹25,000 only. See 4.4 for how each number is built.

Ref	Employee profile	EPF wage	EPS	EPF setting
A	High earner	78,266.50	On	Restricted + NC
B	Mid earner	20,000	On	Restricted + NC
C	High earner	41,360	Off	Restricted + NC

6.2 Test cases

#	Emp	Setting	LOP → paid	EPF wage	EPS wage	EDLI wage	EE PF	A/c 1	EPS	EDLI	ECR NCP
1	A	NC on	0 → 26	19,666.67	19,666.67	19,666.67	2,360	722	1,638	98	0
2	A	NC on	2 → 24	19,666.67	18,355.40	19,666.67	2,360	831	1,529	98	2
3	A	NC on	10 → 16	19,666.67	13,110.80	19,666.67	2,360	1,268	1,092	98	10
4	A	NC on	13 → 13	19,666.67	11,144.13	19,666.67	2,360	1,432	928	98	13
5	B	NC on	2 → 24	16,615.38	16,082.05	16,615.38	1,994	654	1,340	83	2
6	B	NC on	13 → 13	10,000	9,200	10,000	1,200	434	766	50	13
7	A	NC off	2 → 24	18,153.85	18,153.85	18,153.85	2,178	666	1,512	91	2
8	C	EPS off	2 → 24	19,666.67	0	19,666.67	2,360	2,360	0	98	2
9	A	any	26 → 0	—	—	—	0	0	0	0	30 (all wages 0)
10	A	NC on, +2 additional days	2 → 24	21,179.49	19,666.67	19,666.67	2,542	904	1,638	98	0

#	Emp	Setting	LOP → paid	EPF wage	EPS wage	EDLI wage	EE PF	A/c 1	EPS	EDLI	ECR NCP
11	A	Employer unrestricted	2 → 24	72,246 (employer)	18,355.40	19,666.67	depends on employee restriction	12% of 72,246 – 1,529	1,529	98	2
12	A, joined 21-Sep	NC on	18 before joining → 8 paid	24,082.00	10,000	24,082.00	2,890	2,057	833	120	18

Stored wages may differ by ±0.01 (paise rounding). Rupee amounts must match exactly.

How the numbers are built (see 4.2 and 4.4):

Case	Working
1	EPF wage = September blended ceiling: $15,000 \times 16/30 + 25,000 \times 14/30 = 19,666.67$
2–4	EPS wage = each part's reduced ceiling, rounded down, then blended. NCP 2: $14,000 + 9,333 \times 14/30 = 18,355.40$
5	EPF wage: earned $20,000 \times 24/26 = 18,461.54$. 1–16 Sep capped at 15,000; 17–30 Sep 18,461.54 is under 25,000. Blended: $15,000 + 3,461.54 \times 14/30 = 16,615.38$. EPS wage per part: $\min(15,000, 14,000) = 14,000$ and $\min(18,461.54, 23,333) = 18,461.54 \rightarrow 14,000 + 4,461.54 \times 14/30 = 16,082.05$
6	EPF wage: earned $20,000 \times 13/26 = 10,000$, under both caps. EPS wage per part (NCP 13): $\min(10,000, \text{floor}(15,000 \times 17/30) = 8,500) = 8,500$ and $\min(10,000, \text{floor}(25,000 \times 17/30) = 14,166) = 10,000 \rightarrow 8,500 + 1,500 \times 14/30 = 9,200$
7	NC off: each cap $\times 24/26$ (13,846.15 and 23,076.92), blended = 18,153.85
10	Additional days on top of the cap: $15,000 + 15,000 \times 2/26 = 16,153.85$ and $25,000 + 25,000 \times 2/26 = 26,923.08$, blended = 21,179.49. EPS stays at 19,666.67 (NCP 0); the extra goes to EE PF (2,542), A/c 1 (904) and admin (106)
11	Unrestricted employer wage = $78,266.50 \times 24/26 = 72,246$
12	Joined 21-Sep, so only the ₹25,000 ceiling applies (every employed day is after 17-Sep). EPF wage (NC on) = $\min(25,000, 78,266.50 \times 8/26 = 24,082.00) = 24,082.00$. NCP = $26 - 8 = 18$ (working days before joining), EPS contributory calendar days = 12. Reduced EPS ceiling = $\text{floor}(25,000 \times 12/30) = 10,000$, so EPS wage = $\min(24,082, 10,000) = 10,000$ and EPS = 833. EE PF = 12% of 24,082 = 2,890; A/c 1 = $2,890 - 833 = 2,057$. EDLI wage = $\min(24,082, 25,000) = 24,082$, EDLI = 120; admin 120. Employed for 10 calendar days but credited 12 pension days (see section 9)

6.3 Normal month example (no ceiling change)

October 2026: 31 calendar days, ceiling ₹25,000, 27 working days, LOP 2, high earner, restricted + NC on:

NCP	EPF wage	EPS wage	EE PF	EPS	A/c 1	EDLI
2	25,000	$\text{floor}(25,000 \times 29/31) = \mathbf{23,387}$	3,000	1,948	1,052	125

Use the working days of your own payrun and the formulas in section 4 if they differ.

6.4 Checks on every ECR line

1. EPS wage $\leq$ EPF wage.
2. EPS wage $\leq$ ceiling $\times$ (calendar days – NCP) / calendar days. For September use the September blended ceiling for the employee's employed days (19,666.67 for a full month); small differences of a few paise come from rounding each part down first (see 4.4).

3. EDLI wage = lower of the EPF wage (after LOP) and the full ceiling; its ceiling is not reduced for NCP days.
4. A/c 1 + EPS = employer 12% (equals EE PF when employee and employer are both restricted).
5. NCP = calendar days only when nothing is paid.
6. Net pay and employee PF are the same as before this change for the same LOP.

6.5 Optional / dev-assisted checks

Check	Expected
26th-to-25th pay cycle	Calendar days come from the month of the payroll date (cycle end)
Old payrun (calculated before release) with LOP, not recalculated	ECR NCP 0, EPS wage as stored
Same payrun after recalculation	Real NCP, reduced EPS wage
Old September 2026 payrun (EPS on the unreduced 15k/25k blend) with LOP	ECR NCP 0, EPS wage as stored, warning logged
New September 2026 payrun for an employee who joined after 17-Sep, with LOP	Real NCP (ceiling blended over employed days)
Retro salary revision (arrears) for a past month with LOP	EPS arrear capped at that month's reduced ceiling
Full & final payrun in the month	Not included in the monthly ECR

7. Expected behaviour (not bugs)

Observation	Explanation
With NC off or for low earners, EPS wage = EPF wage even with LOP	Paid-day proration already brings the wage below the reduced ceiling
Half-day LOP shows 1 NCP day; a quarter day shows 0	ECR accepts whole days only; NCP is rounded to the nearest day, halves up
Mid-month joiner / leaver shows NCP for days before joining or after exit	Wages are not due for those days (EPFO's NCP definition)
DOJ / DOE are not in the ECR file	They are marked on the EPFO portal (UAN joining details and "mark exit"), not in the ECR text file
Old payrun with LOP shows NCP 0	Not recalculated after release; recalculate to apply the new rules
EPF wage and EE PF don't change with LOP for NC-on high earners	NC keeps the capped wage until earnings fall below the cap; only EPS changes

8. Scope

- **Monthly ECR only.** The ECR report covers regular monthly payruns. Full & final payruns are not included in the monthly ECR (including them was a bug); full & final filing is handled separately.
- **Out of scope:** automatically recalculating payruns calculated before this release. Recalculate them before their ECR is filed (see 4.5).

9. Interpretations (no explicit EPFO rule)

These are product decisions. EPFO has not published a rule for them; they are documented so they are not mistaken for statutory requirements.

Decision	What we do
EPS ceiling reduced for NCP days	EPS wage capped at ceiling $\times$ (calendar days – NCP) / calendar days; the surplus goes to EPF A/c 1
NCP counted in working days	NCP = working days – paid days, consistent with how wages are paid; then subtracted from calendar days for the EPS cap
Days outside employment are NCP	Working days before the date of joining or after the date of exit are counted as NCP (wages are not due for them). Counted in working days, so a joiner's EPS contributory calendar days can be a few days more than their employed calendar days (weekends before joining are not added)
NCP rounding	Nearest whole day, halves rounded up (EPFO only requires whole days)
Additional earning days	Count as paid days (reduce NCP) and are contributed on top of the cap

No open questions.

10. FAQ

Does the employee's take-home pay change? No. Employee PF and net pay are unchanged. Only the employer split between EPS (pension) and EPF A/c 1 changes.

Does the employer pay more or less? Neither. The employer's total 12% is unchanged; a smaller EPS share moves into A/c 1.

Why is the EDLI ceiling not reduced for NCP days? EDLI is life-insurance cover and is not linked to pensionable service days. The EDLI wage still follows the EPF wage after LOP; only its ceiling stays at the full ₹15,000 / ₹25,000.

What if an employee had no pay at all? The ECR line shows NCP equal to the days in the month and all wages 0.

Do additional earning days matter? Yes. They count as paid days, so they reduce (or cancel) NCP.